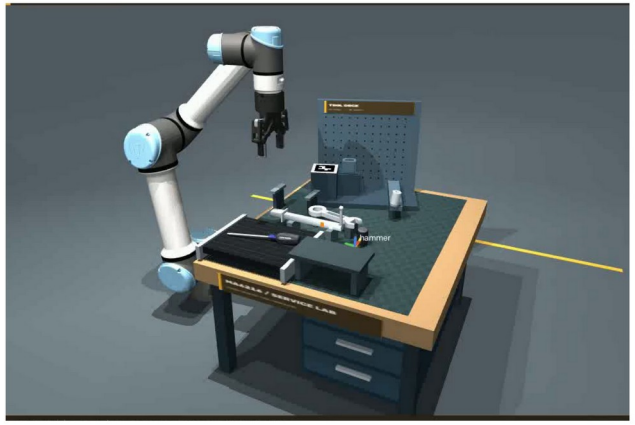
# Calibrated 6D vision for sequential tool manipulation



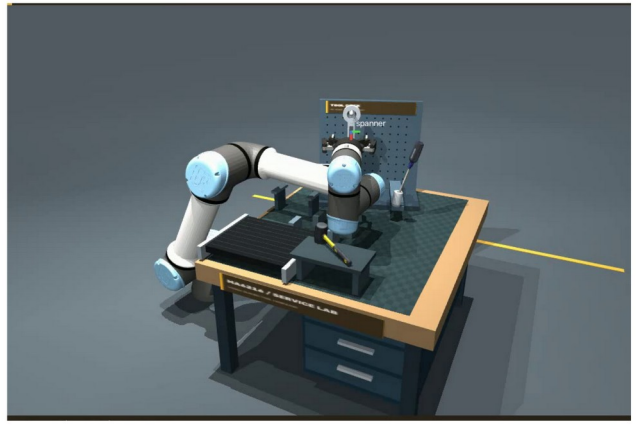
Problem & scope

**a** Initial workcell



t = 0.00 s | object pickup stations

**b** Released final state



t = 195.00 s | four acceptance checks passed

## Study objective

UR5e + Robotiq 2F-85 in MuJoCo

RGB-D reconstruction + FoundationPose

Does execution-period pose feedback improve complete-task reliability?

| Operation | Required relationship | Acceptance |
|---|---|---|
| Hammer return | Tool-to-tray pose | Released support |
| Moving screwdriver | Shaft-to-tilted-socket axis | Insertion + release |
| Independent rack | Frame-to-base seating | Stable fixture |
| Spanner regrasp / hang | New grasp; hole-to-pin alignment | Released hanging |

Scope: one continuous four-operation scene. Contribution: independent integration and evaluation of existing methods.

# System architecture and technical selection

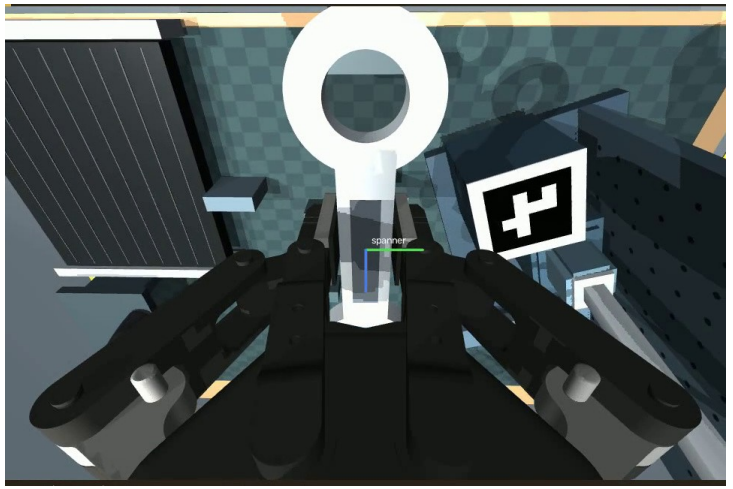


Methods

**RGB-D**
Estimated K
→
**Localization**
Depth foreground
→
**6D pose**
FoundationPose
→
**Robot target**
Calibrated chain
→
**Execution**
IK + control

Reobserve → update pose → bounded correction; simulation truth enters downstream scoring only.

| Method / component | Output / role | Selection boundary |
|---|---|---|
| FoundationPose | Object-frame 6D pose | Selected; reconstructed RGB-D references |
| MegaPose | Model-based 6D pose | Reviewed; no external benchmark |
| GraspNet | Grasp candidates | Does not alone specify assembly pose |
| MuJoCo / OpenCV | Contact simulation / calibration | Reuse platform; inspectable interfaces |



Model-free route: 24 calibrated RGB-D reference views → object-specific reconstruction → registration / tracking.

References and pretrained networks are still required; synthetic reference calibration remains an assumption.

# Camera calibration: estimation and independent validation



Methods

**a** Checkerboard observations

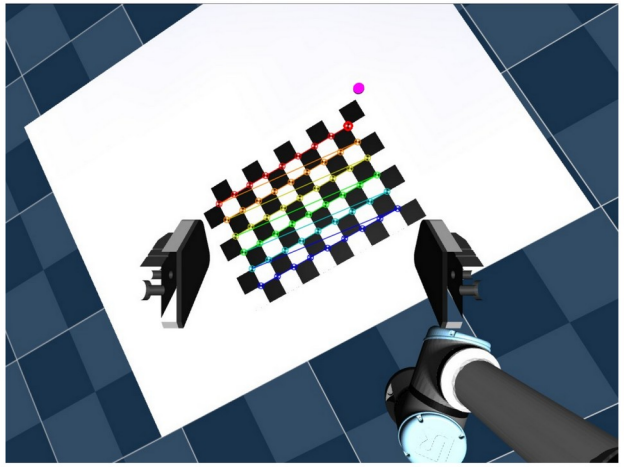
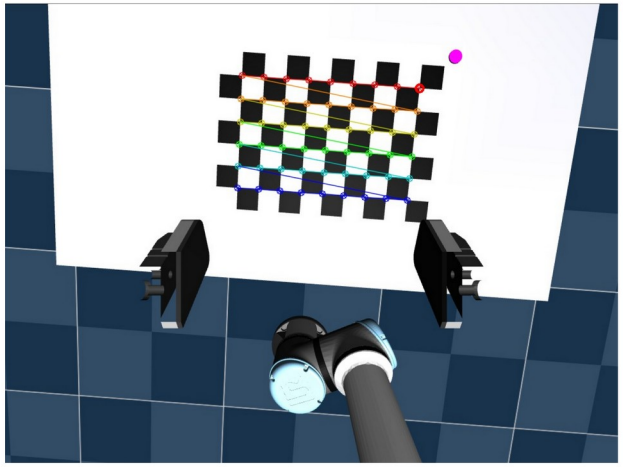


Image-based corner detection

**b** Held-out hand–eye view



PnP + robot FK; frozen validation chain

| Procedure | Train / validation | Validation result |
|---|---|---|
| Wrist intrinsics | 24 / 8 views | 0.02177 px RMSE |
| Wrist hand–eye | 24 / 10 views | 0.852 mm / 0.0227° |
| Frozen chain | K, X and Y fixed | 0.05497 px |

## Estimation protocol

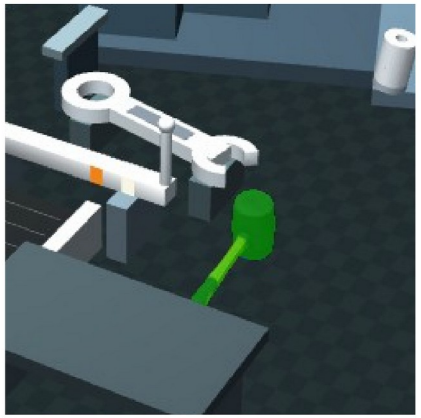
Checkerboard pixels → intrinsic calibration → image PnP + robot FK → AX = XB → training-only refinement.

Simulator K and camera pose are validation references only. Rejected images and held-out views are retained.
The known distortion-free pinhole model explains the small residuals; these are not real-camera accuracy claims.

# Localization: image masks, evaluation unit and assumptions

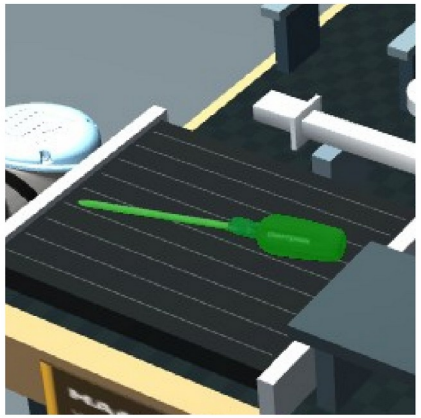


**a** Hammer



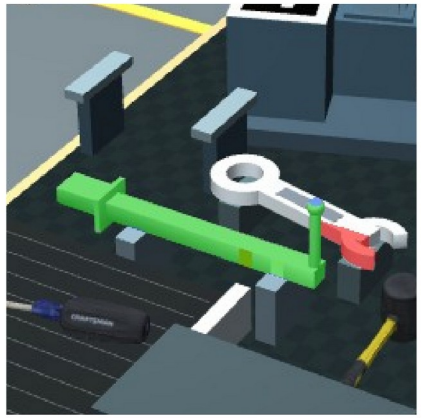
Nominal scene | seed 500

**b** Screwdriver



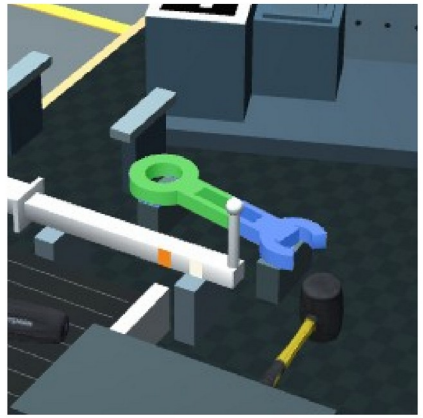
Nominal scene | seed 500

**c** Rack frame



Nominal scene | seed 500

**d** Spanner



Nominal scene | seed 500

| Detector | Independent evaluation | Boundary |
|---|---|---|
| Depth foreground subtraction<br>Known pickup regions | 50 scenes / 200 queries<br>10 scenes per tool–condition cell | Matched empty backgrounds<br>Restricted occlusion / noise |
| Predictions saved first | 160/160 positive queries: IoU ≥ 0.5<br>0/40 absent-target false positives | Wilson and scene-bootstrap intervals<br>Not 200 robot executions |

Nominal seed 500. Green: overlap; blue: missed visible target; red: extra foreground. Crops preserve original overlays.

# 6D pose estimation and coordinate transformation





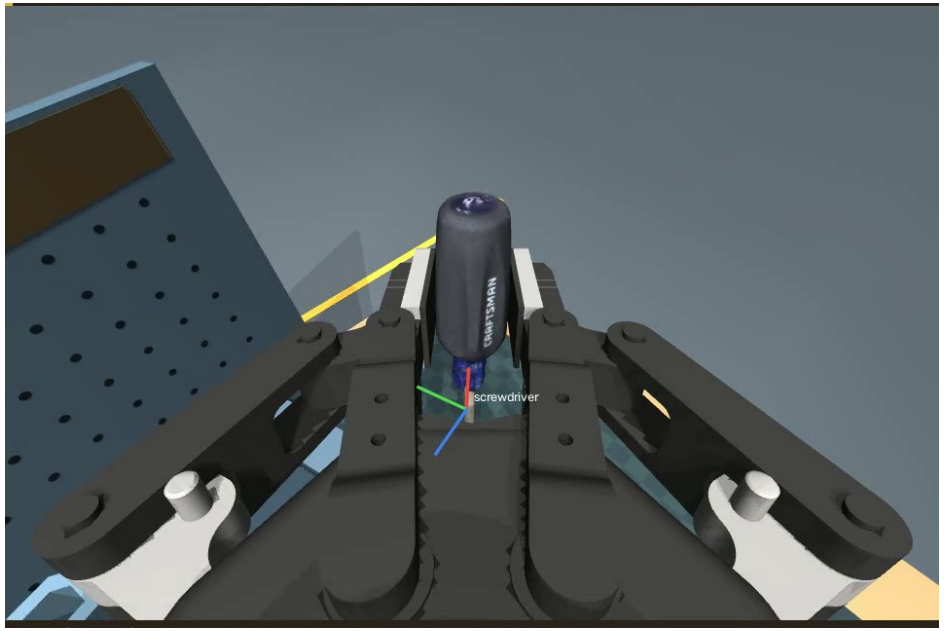
$$T_{BO} = T_{BG} \cdot T_{GC} \cdot T_{CO}$$

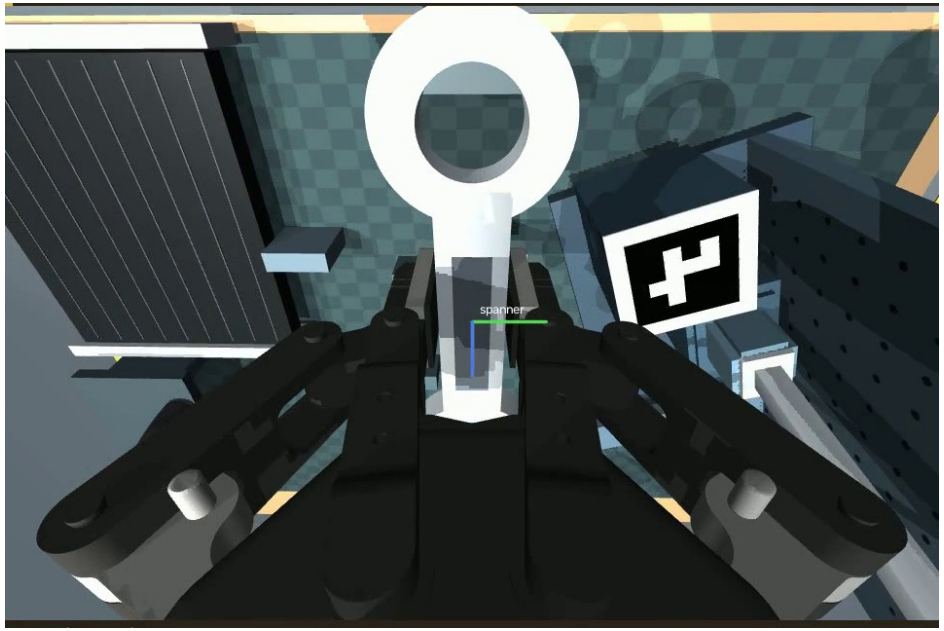
B: robot base  G: gripper  C: wrist camera  O: object

T_AB maps coordinates from B into A.

FK supplies T_BG; calibration estimates T_GC.

FoundationPose supplies T_CO from visual observations.




Task target: estimated object pose + task-specific geometric offset.

Regrasp changes the object-to-gripper relationship; observe again before generating the next target.

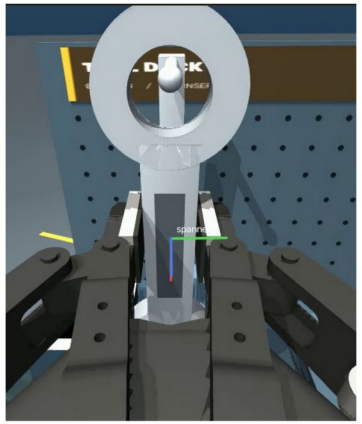
# Visual feedback: modes, residuals and correction policy





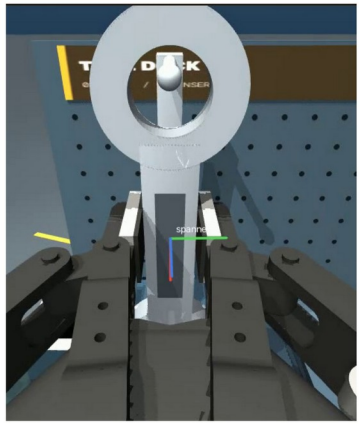
| Mode | Execution-period observation | Common components |
|---|---|---|
| Open loop | Initial acquisition / feed prediction only | Same estimator, calibration and guards |
| Staged | Selected task-transition reobservations | Same physical acceptance |
| Iterative | Repeated precontact pose correction | Staged observations retained |

**a** Before corrective action



Visual angular residual: 1.60°

**b** After correction and reobservation



Visual angular residual: 0.74°

$$e_p = p_{des} - p_{est}$$

$$\Delta R = R_{des}\, R_{est}^{T}$$

Cap: 4 mm / 2° per step

Converged: two observations within 1.5 mm / 1.5°

Discrete precontact control; no stability proof.

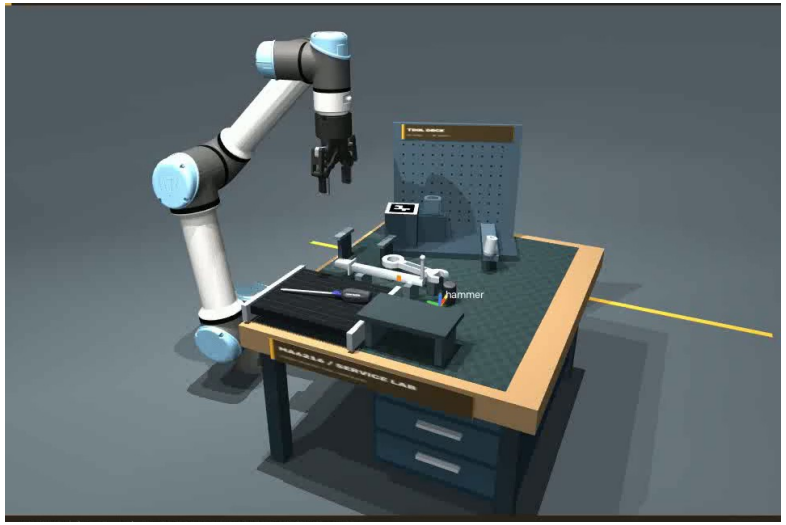
Correction images: separate perturbed integration run, one logged correction; 1.60° → 0.74° are visual residuals, not true geometric errors.

# Experimental design and metric definitions



Evaluation protocol

| Factor | Levels / unit | Control |
|---|---|---|
| Feedback | Open loop / staged / iterative | Same pose estimator and camera calibration |
| Disturbance | None / alignment force–torque pulse | Delivery logged; early failures retained |
| Randomization | 8 seeds per cell; 48 attempts | Initial states paired within seed |
| Primary outcome | All four final physical checks | Geometry + support + release + stability |



## Statistical reporting

Success: Wilson 95% confidence intervals.

Paired comparisons: exact tests with Holm adjustment.

Continuous metrics: trial-level summaries and bootstrap intervals.

Seeds 400–407 are reused: paired replication, not a new holdout.

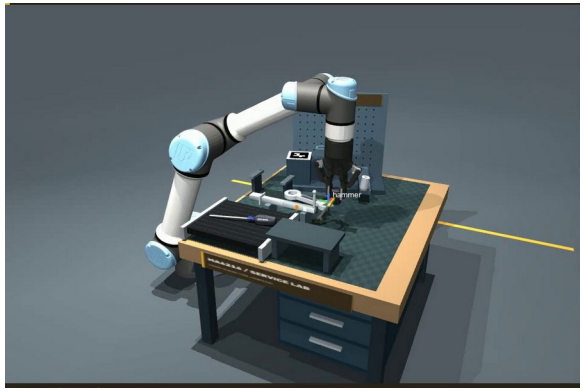
No task failures are removed or pooled with historical studies.

# Execution evidence I: tool return and tilted insertion

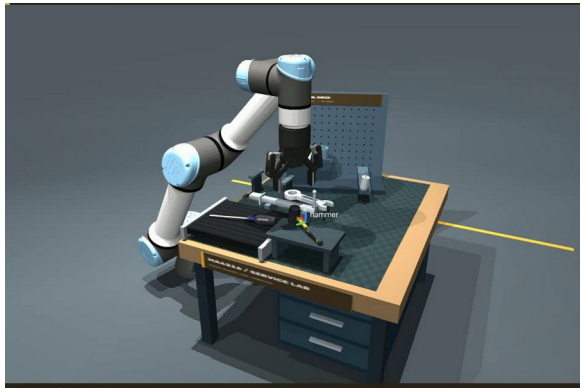


Representative successful run

a Lift / 13.60 s

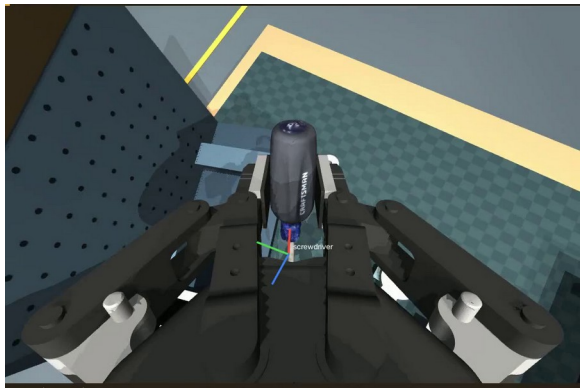


b Released placement / 25.00 s

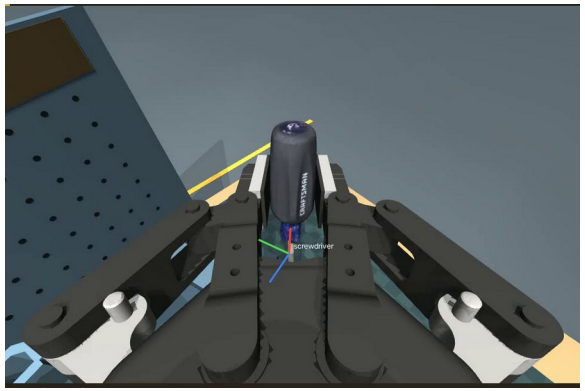


a–b Hammer: held transport → released tray placement. Support and stability determine acceptance.

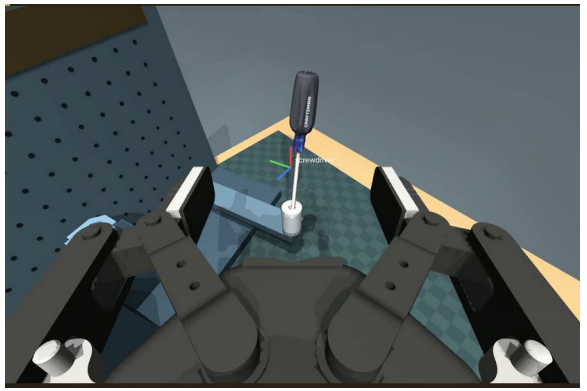
a Reorientation / 50.00 s



b Insertion / 60.00 s



c Released / 70.00 s



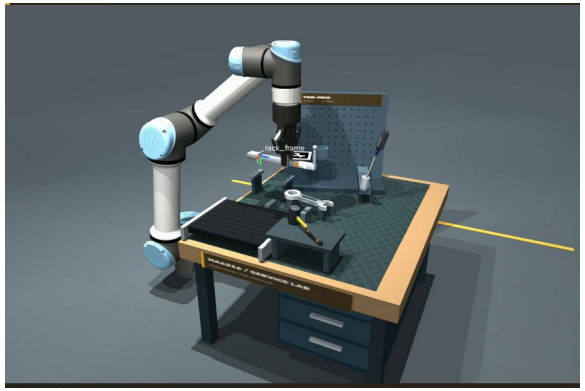
a–c Screwdriver: reorientation → guarded insertion → released placement. Initial moving-target acquisition precedes these frames.

# Execution evidence II: rack installation and physical regrasp

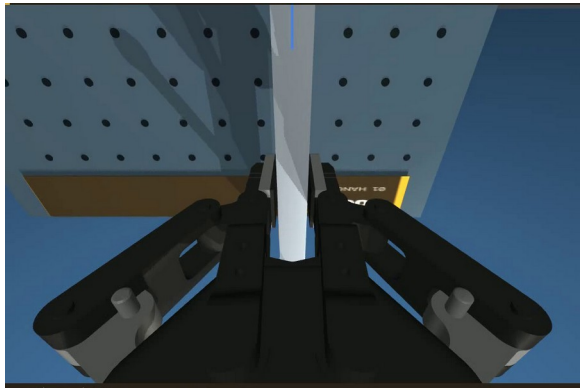


Representative successful run

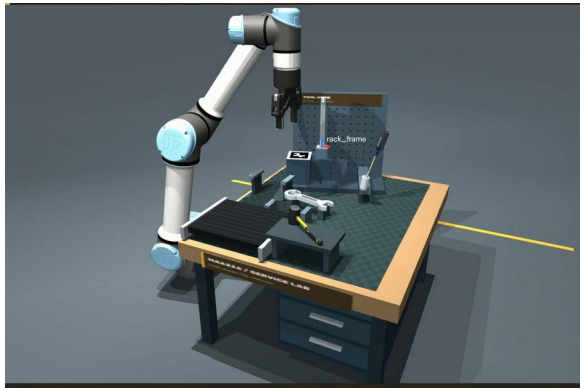
a Frame pickup / 91.00 s



b Seating / 112.00 s

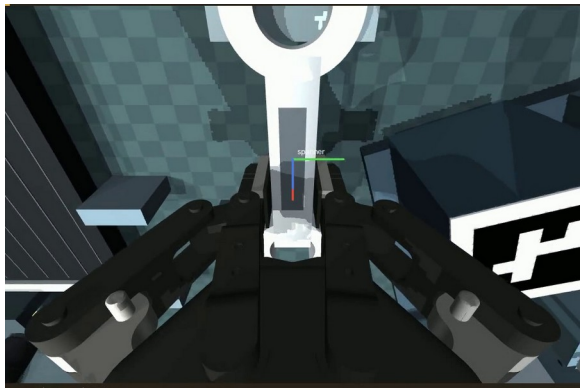


c Released fixture / 134.40 s



a–c Rack: independent frame → seating → released fixture; relocalize before the next operation.

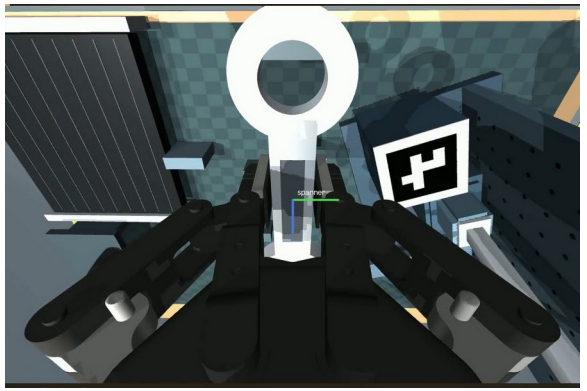
a Supported / 147.00 s



b Finger clearance / 150.40 s



c New grasp / 158.20 s



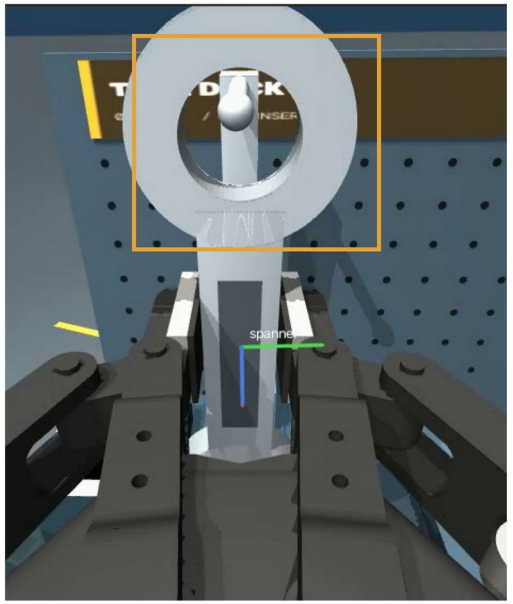
a–c Spanner: supported placement → finger clearance → new grasp after observation. One continuous original trajectory.

# Hole-over-pin placement and terminal acceptance



Representative successful run

**a** Precontact alignment



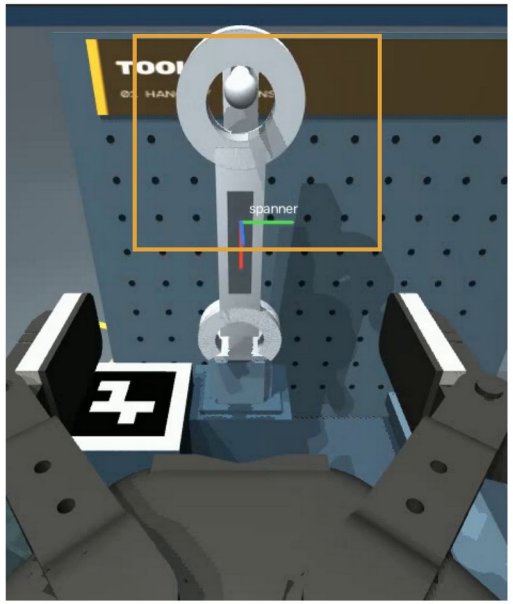

t = 171.00 s

**b** Hole-over-pin threading




t = 179.00 s

**c** Released hanging state




t = 195.00 s

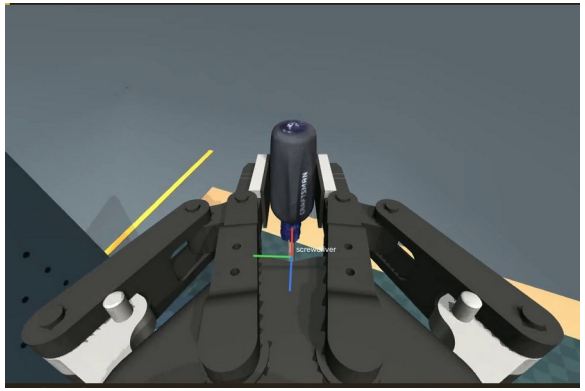
| Visual relationship | Controller / physical evidence | Interpretation |
|---|---|---|
| Visual alignment | Thread → seat → release | Explanatory ROI outlines |
| Final support | Four terminal checks passed | Single feasible trajectory |
| Passive stability | 30 s stable after release | Separate diagnostic |

# Paired qualitative comparison and failure boundary

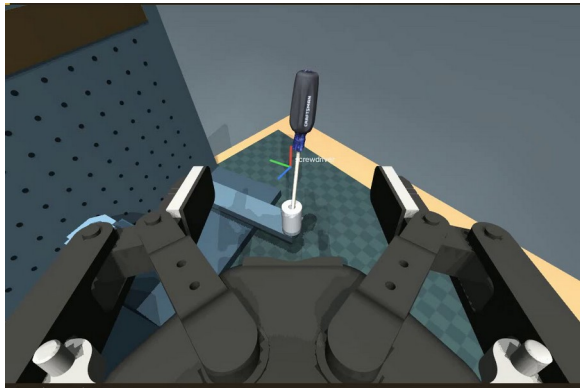


Controlled examples

a Open loop / guard stop / 57.96 s

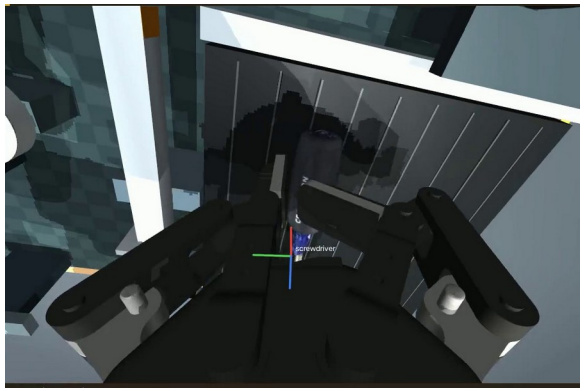


b Iterative / released / 70.00 s

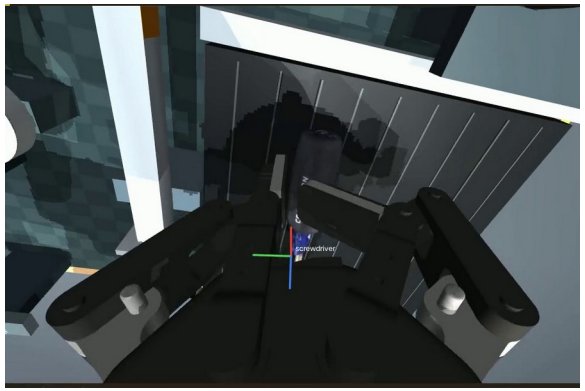


Seed 400, no disturbance: open-loop insertion guard stop vs iterative released placement. Different endpoint times.

a Staged / pickup stop / 40.24 s



b Iterative / pickup stop / 40.24 s

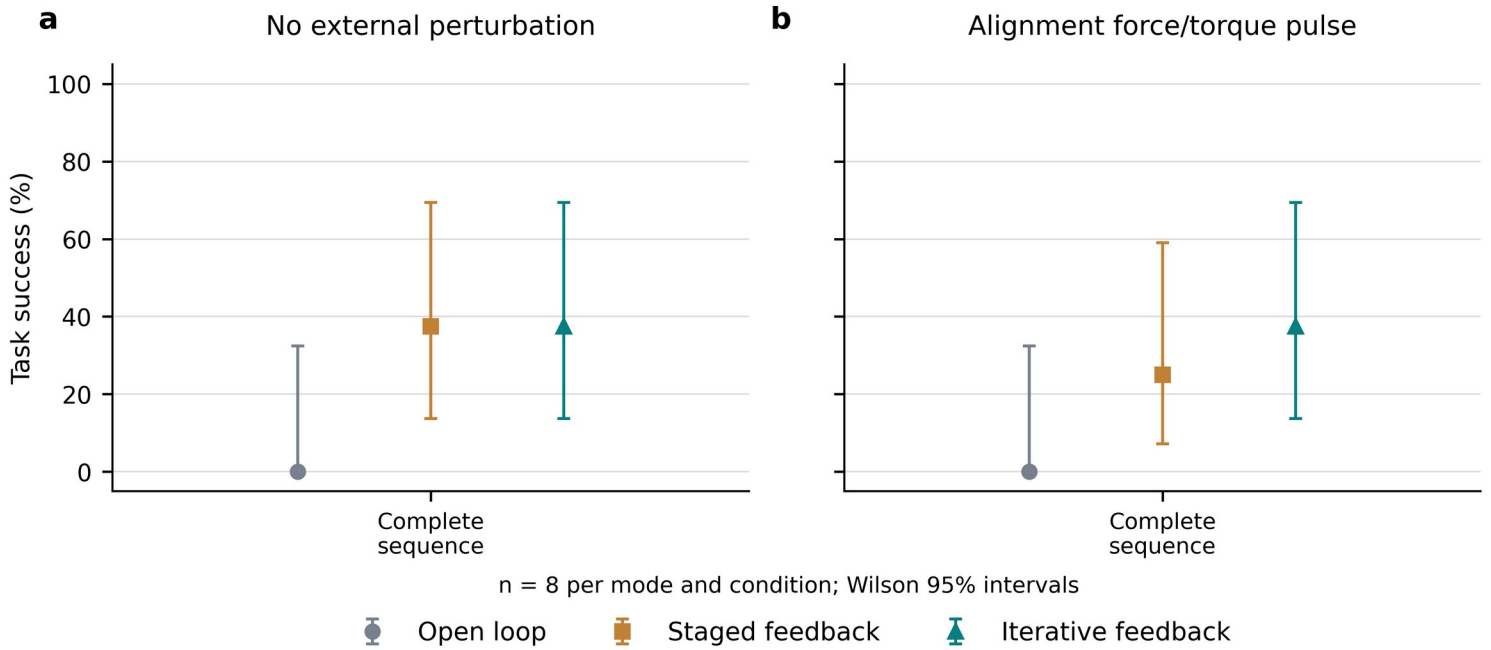


Seed 401: both staged and iterative modes stop at gripper closure. Contact causes come from logs, not visual inference.

# Whole-sequence results and statistical uncertainty



Quantitative results




| Mode | None | Pulse |
|---|---|---|
| Open loop | 0/8 | 0/8 |
| Staged | 3/8 | 2/8 |
| Iterative | 3/8 | 3/8 |

11/48 complete successes.
All four adjusted paired p-values: 1.0.
No statistically established superiority or equivalence in this sample.

## Interpretation

The selected successful trajectory demonstrates feasibility. Low complete-sequence reliability and wide intervals limit conclusions about feedback benefit. Early failures remain part of the system-level outcome.

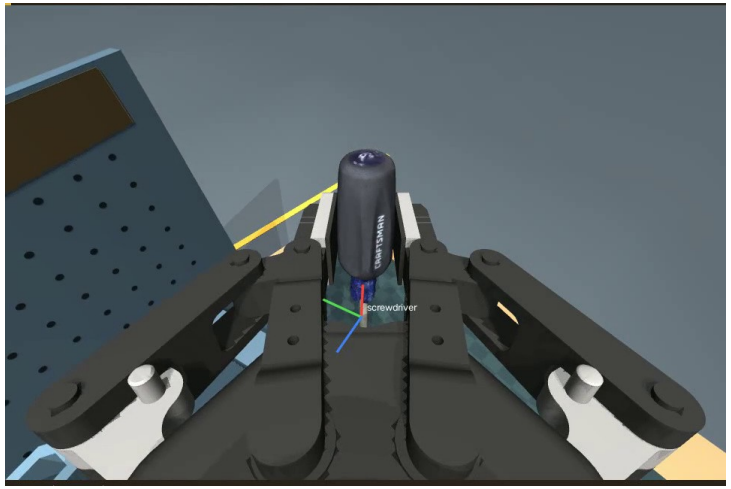
# Diagnostic results: pose accuracy and physical failures

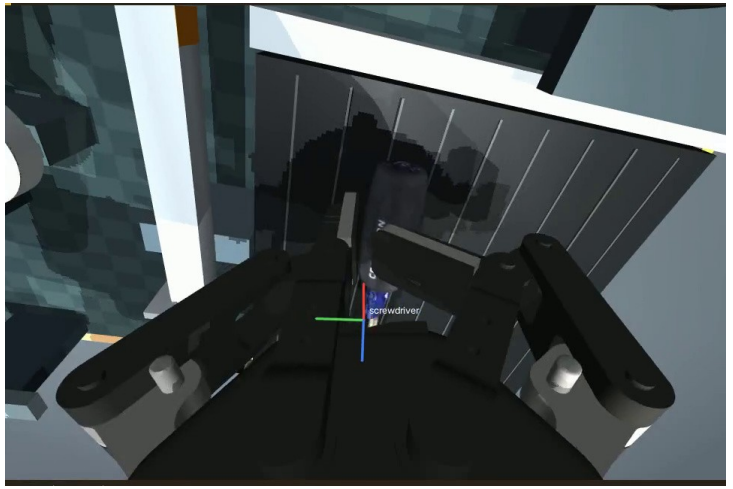


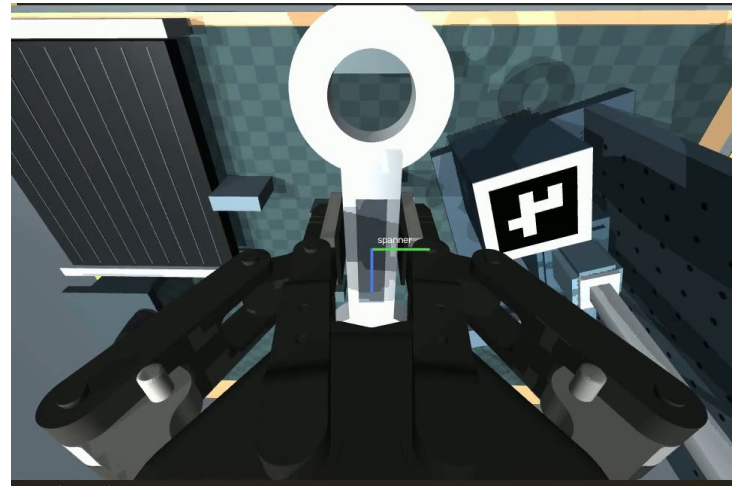
Error decomposition

| Initial pose metric | Mean |
|---|---|
| Translation error | 2.60 mm |
| Rotation error | 1.44° |
| ADD / ADD-S | 3.17 / 2.29 mm |

| Failure category | Attempts |
|---|---|
| Moving-pickup contact guard | 24 |
| Hammer approach planning | 6 |
| Open-loop insertion guard | 6 |
| Bilateral spanner grasp loss | 1 |



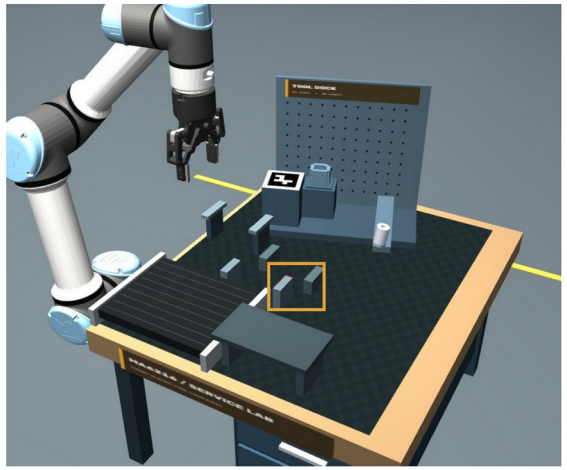

Different endpoints: initial pose error, TCP tracking error, final placement and task success. 500 Hz physics is not visual feedback frequency.

# Generalization boundary: background-dependent localization



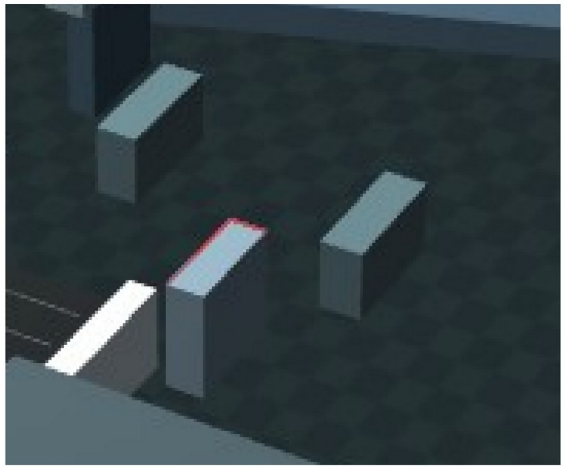


**a** Target-absent scene



Separate background-mismatch diagnosis

**b** Magnified false-positive region



Red pixels: predicted foreground without target

## Observed limitation

Absent rack target, but mismatched background geometry produces a false-positive region.

The zoom is from the same original image.

| Assumption | Impact on interpretation |
|---|---|
| Known pickup stations and paired backgrounds | Does not establish arbitrary-clutter recognition |
| Synthetic scans, ideal FK / depth, rigid mounts | Does not establish real-camera or hardware reliability |
| Small reused-seed sample | Broader unseen-scene trials remain necessary |

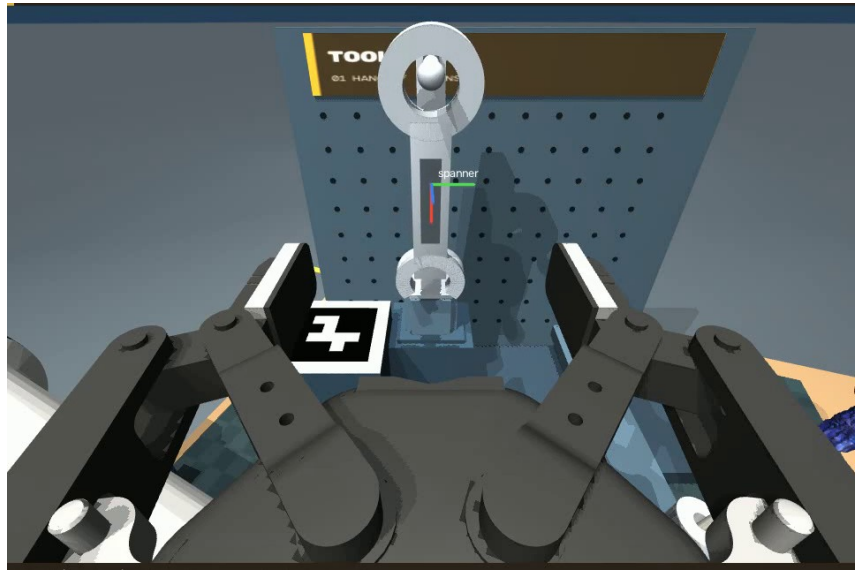
Separate development diagnosis; not pooled into the 50-scene localization audit or the 48 robot attempts.

# Conclusions and priorities for further experiments





| Supported by the evidence | Not established |
| --- | --- |
| Independent calibration integrated into visual control | Real-robot accuracy and reliability |
| Physical four-operation sequence with regrasp | High success rate across arbitrary layouts |
| Auditable internal feedback comparison | Statistical superiority or external-algorithm ranking |




## Next experiments

1. Improve pickup planning and bilateral-contact reliability.
2. Test segmentation under changed backgrounds and clutter.
3. Evaluate pose bias and observation scheduling.
4. Use unseen seeds and matched external-estimator baselines.
5. Validate the calibrated pipeline on physical hardware.

# References and reproducibility



Appendix

Selected methods

Wen et al. FoundationPose. CVPR, 2024.

Zhang. A flexible new technique for camera calibration. TPAMI, 2000.

OpenCV. Camera and hand–eye calibration documentation.

Chaumette & Hutchinson. Visual servo control, Part I. RAM, 2006.

Todorov et al. MuJoCo. IROS, 2012.

Reviewed alternatives

Labbé et al. MegaPose. arXiv:2212.06870, 2022.

Fang et al. GraspNet-1Billion. CVPR, 2020.

Repository evidence

Calibration images + robot FK

Predicted and evaluation masks

48 complete trial summaries

Original successful and failed replays

12 annotated image plates (PDF / SVG)

Frame indices, crop boxes and source hashes

Full references: docs/LITERATURE_AND_SELECTION.md

Experiment report: visual-feedback-calibrated-v1

Simulation images illustrate observed behavior; quantitative conclusions use all predefined attempted trials.